

ELECTROMAGNETIC WAVES

Important formulas and concepts directly from NCERT-

- A time varying electric field generates magnetic field.
- The most important prediction from Maxwell's equation is existence of electromagnetic(EM) waves, which are time varying electric and magnetic fields that propagate in space.
- Speed of EM waves is nearly equal to speed of light, this makes a conclusion that 'Light is an EM wave'.

❖ Displacement current-

- Maxwell found an inconsistency in the ampere's law and suggested the existence of an additional current, the displacement current to remove this inconsistency.
- This displacement current is due to time varying electric field and given by,

$$i_D = \epsilon_0 d\phi_E / dt$$

- This current acts as a source of magnetic field in exactly the same way as conduction current.

Conduction current – current due to flow of charges.

- Ampere maxwell law,

$$\begin{aligned}\oint \mathbf{B} \cdot d\mathbf{l} &= \mu_0 i_C + \mu_0 i_D \\ &= \mu_0 i_C + \mu_0 \epsilon_0 d\phi_E / dt\end{aligned}$$

MAXWELL'S EQUATIONS

1. $\oint \mathbf{E} \cdot d\mathbf{A} = Q / \epsilon_0$ (Gauss's Law for electricity)
2. $\oint \mathbf{B} \cdot d\mathbf{A} = 0$ (Gauss's Law for magnetism)
3. $\oint \mathbf{E} \cdot d\mathbf{l} = \frac{-d\phi_B}{dt}$ (Faraday's Law)
4. $\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 i_c + \mu_0 \epsilon_0 \frac{d\phi_E}{dt}$ (Ampere – Maxwell Law)

❖ Electromagnetic waves-

- Accelerated charges produce electromagnetic waves.
- Electric and magnetic fields in an EM wave are perpendicular to each other and to direction of propagation.
- EM wave propagating in z direction, electric field is along the x axis and magnetic field is along the y axis.

$$E_x = E_0 \sin(kz - \omega t)$$

$$B_y = B_0 \sin(kz - \omega t)$$

Here, $k = 2\pi/\lambda$

Where, ω = angular frequency and

λ = wavelength

k is the magnitude of wave vector and its direction describes the direction of propagation of wave.

- Speed of propagation of wave,

$$c = \omega/k$$

$$c = 1/\sqrt{\mu_0 \epsilon_0}$$

In terms of frequency, $v\lambda = c$

$$(\omega/2\pi) (2\pi/k) = c$$

- Relation b/w magnetic and electric field in an EM wave,

$$B_0 = E_0/c$$

- In a material medium of permeability μ and permittivity ϵ , the velocity of light becomes,

$$v = 1/\sqrt{\mu\epsilon}$$

thus, velocity of light depends on electric and magnetic properties of medium.

- The velocity of EM wave is independent of wavelength.

❖ Electromagnetic spectrum-

- The classification of EM waves according to frequency is the electromagnetic spectrum.
- There is no sharp division b/w one kind of wave and the next. The classification is based roughly on how the waves are produced or detected.

1) Radio waves-

- Used in radio and television communication systems.
- Cellular phones use radio waves to transmit voice communication in the ultrahigh frequency band.

2) Micro waves-

- Suitable for radar systems used in aircraft navigation.
- Radar also provides the basis for the speed guns used to time fast balls, tennis services and automobiles.
- In microwave ovens, 'the frequency of microwaves is selected to match the resonant frequency of water molecules so that energy from the waves is transferred efficiently to the kinetic energy of the molecules. This raises the temperature of any food containing water.'

3) Infrared waves-

- Produced by **hot bodies and molecules.**
- Sometimes referred to as 'heat waves'.
- Water molecules present in most materials easily absorb infrared molecules (other molecules like CO_2 , NH_3 also absorb infrared waves)
- Used in physical therapy.

- It also plays important role in maintaining earth's warmth temp. through greenhouse effect.
- Also used in earth satellites, military purposes and to observe growth of crops.
- Used in **remote switches of systems such as TV sets, video recorders and hi-fi systems.**

4) Visible rays-

- It is the part of spectrum that is detected by the **human eye**.
- Different animals sensitive to different wavelengths, for example **snakes can detect infrared waves, and 'visible' range of many insects extends well into the ultraviolet.**

5) Ultraviolet rays-

- Produced by **special lamps and very hot bodies.**
- Exposure to UV radiation induces the production of more melanin, causing tanning of the skin.
- UV radiation is absorbed by ordinary glass. Hence, one can't get tanned through glass windows.
- Can be used in applications such as **LASIK**(Laserassisted in situ keratomileusis) eye surgery.
- Used to kill germs in water purifiers.
- Ozone layer in atmosphere plays a protective role by absorbing UV rays, hence its depletion by CFCs such as freons is a matter of concern.

6) X-rays-

- Used as a **diagnostic tool in medicine** and as a **treatment for certain forms of cancer**.
- Because X-rays damage or destroy living tissues and organisms, care must be taken to avoid unnecessary or over exposure .

7) Gamma rays-

- Produced in **nuclear reactions** and also emitted by **radioactive nuclei**.
- Used in medicine to destroy cancer cells.

TABLE 8.1 DIFFERENT TYPES OF ELECTROMAGNETIC WAVES

Type	Wavelength range	Production	Detection
Radio	> 0.1 m	Rapid acceleration and decelerations of electrons in aerials	Receiver's aerials
Microwave	0.1m to 1 mm	Klystron valve or magnetron valve	Point contact diodes
Infra-red	1mm to 700 nm	Vibration of atoms and molecules	Thermopiles Bolometer, Infrared photographic film
Light	700 nm to 400 nm	Electrons in atoms emit light when they move from one energy level to a lower energy level	The eye Photocells Photographic film
Ultraviolet	400 nm to 1nm	Inner shell electrons in atoms moving from one energy level to a lower level	Photocells Photographic film
X-rays	1nm to 10^{-3} nm	X-ray tubes or inner shell electrons	Photographic film Geiger tubes Ionisation chamber
Gamma rays	$<10^{-3}$ nm	Radioactive decay of the nucleus	-do-

SUMMARY

1. Maxwell found an inconsistency in the Ampere's law and suggested the existence of an additional current, called displacement current, to remove this inconsistency. This displacement current is due to time-varying electric field and is given by

$$i_d = \epsilon_0 \frac{d\Phi_E}{dt}$$

and acts as a source of magnetic field in exactly the same way as conduction current.

2. An accelerating charge produces electromagnetic waves. An electric charge oscillating harmonically with frequency ν , produces electromagnetic waves of the same frequency ν . An electric dipole is a basic source of electromagnetic waves.
3. Electromagnetic waves with wavelength of the order of a few metres were first produced and detected in the laboratory by Hertz in 1887. He thus verified a basic prediction of Maxwell's equations.
4. Electric and magnetic fields oscillate sinusoidally in space and time in an electromagnetic wave. The oscillating electric and magnetic fields, $\mathbf{E}$ and $\mathbf{B}$ are perpendicular to each other, and to the direction of propagation of the electromagnetic wave. For a wave of frequency ν , wavelength λ , propagating along z-direction, we have

$$E = E_x(t) = E_0 \sin(kz - \omega t)$$

$$= E_0 \sin \left[2\pi \left(\frac{z}{\lambda} - \nu t \right) \right] = E_0 \sin \left[2\pi \left(\frac{z}{\lambda} - \frac{t}{T} \right) \right]$$

$$B = B_y(t) = B_0 \sin(kz - \omega t)$$

$$= B_0 \sin \left[2\pi \left(\frac{z}{\lambda} - \nu t \right) \right] = B_0 \sin \left[2\pi \left(\frac{z}{\lambda} - \frac{t}{T} \right) \right]$$

They are related by $E_0/B_0 = c$.

5. The speed c of electromagnetic wave in vacuum is related to μ_0 and ϵ_0 (the free space permeability and permittivity constants) as follows:

$c = 1/\sqrt{\mu_0 \epsilon_0}$. The value of c equals the speed of light obtained from optical measurements.

Light is an electromagnetic wave; c is, therefore, also the speed of light. Electromagnetic waves other than light also have the same velocity c in free space.

The speed of light, or of electromagnetic waves in a material medium is given by $v = 1/\sqrt{\mu \epsilon}$

where μ is the permeability of the medium and ϵ its permittivity.

6. The spectrum of electromagnetic waves stretches, in principle, over an infinite range of wavelengths. Different regions are known by different names; γ -rays, X-rays, ultraviolet rays, visible rays, infrared rays, microwaves and radio waves in order of increasing wavelength from $10^{-2} \text{ \AA}$ or 10^{-12} m to 10^6 m .

They interact with matter via their electric and magnetic fields which set in oscillation charges present in all matter. The detailed interaction and so the mechanism of absorption, scattering, etc., depend on the wavelength of the electromagnetic wave, and the nature of the atoms and molecules in the medium.

POINTS TO PONDER

1. The basic difference between various types of electromagnetic waves lies in their wavelengths or frequencies since all of them travel through vacuum with the same speed. Consequently, the waves differ considerably in their mode of interaction with matter.
2. Accelerated charged particles radiate electromagnetic waves. The wavelength of the electromagnetic wave is often correlated with the characteristic size of the system that radiates. Thus, gamma radiation, having wavelength of 10^{-14} m to 10^{-15} m, typically originate from an atomic nucleus. X-rays are emitted from heavy atoms. Radio waves are produced by accelerating electrons in a circuit. A transmitting antenna can most efficiently radiate waves having a wavelength of about the same size as the antenna. Visible radiation emitted by atoms is, however, much longer in wavelength than atomic size.
3. Infrared waves, with frequencies lower than those of visible light, vibrate not only the electrons, but entire atoms or molecules of a substance. This vibration increases the internal energy and consequently, the temperature of the substance. This is why infrared waves are often called *heat waves*.
4. The centre of sensitivity of our eyes coincides with the centre of the wavelength distribution of the sun. It is because humans have evolved with visions most sensitive to the strongest wavelengths from the sun.

Dr. Aman Tilak
MBBS, AIIMS Delhi